

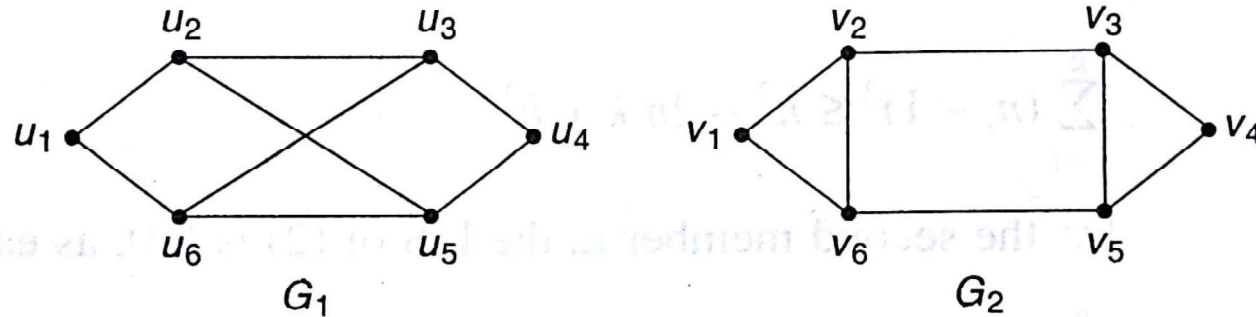
Circuits and Isomorphism

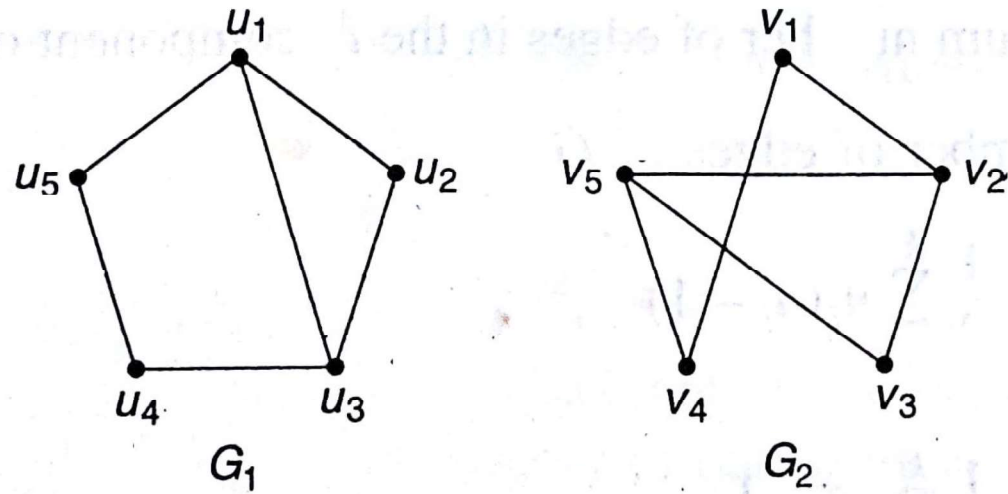
Apart from the three invariants of two isomorphic graphs already discussed, namely number of vertices, number of edges and degrees of corresponding vertices, we have one more invariant of isomorphic graphs.

If two graphs are isomorphic, they will contain circuits of the same length k , where $k > 2$.

If this invariant condition is not satisfied then the two graphs will not be isomorphic.

For example, the two graphs G_1 and G_2 given in Fig. have 6 vertices each, 8 edges each, 4 vertices of degree 3 and 2 vertices of degrees 2. still they are not isomorphic, because G_2 has a circuit of length 3, namely, $v_1 - v_2 - v_3 - v_1$, whereas G_1 has no circuit of length 3.





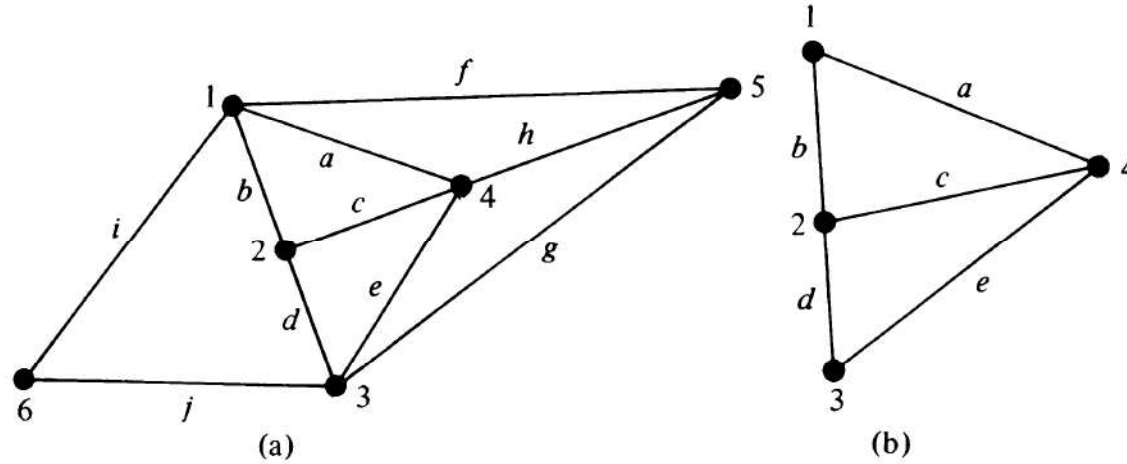
The two graphs G_1 and G_2 given in Fig. satisfy the usual (three) invariant conditions. We also note they have got circuits of length 5 which pass through all vertices, namely, $u_1 - u_2 - u_3 - u_4 - u_5 - u_1$ and $v_5 - v_3 - v_2 - v_1 - v_4 - v_5$.

Also we can verify that their adjacency matrix is same.

$$A_{G_1} \equiv \begin{matrix} & \begin{matrix} u_1 & u_2 & u_3 & u_4 & u_5 \end{matrix} \\ \begin{matrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}; \quad A_{G_2} \equiv \begin{matrix} & \begin{matrix} v_5 & v_3 & v_2 & v_1 & v_4 \end{matrix} \\ \begin{matrix} v_5 \\ v_3 \\ v_2 \\ v_1 \\ v_4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

SUBGRAPHS

A graph g is said to be a *subgraph* of a graph G if all the vertices and all the edges of g are in G , and each edge of g has the same end vertices in g as in G .

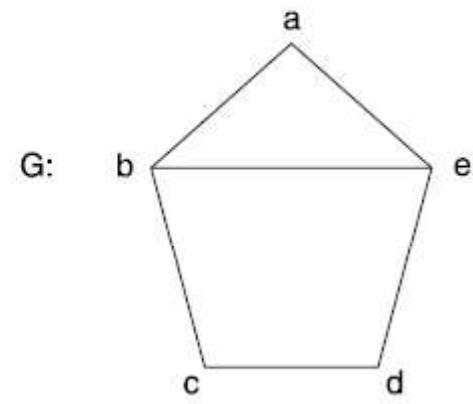
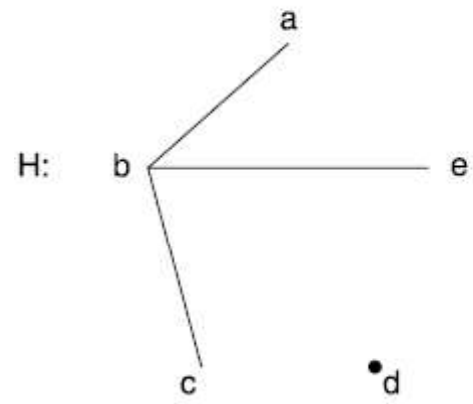


Graph (a) and one of its subgraphs (b).

The following observations can be made immediately:

1. Every graph is its own subgraph.
2. A subgraph of a subgraph of G is a subgraph of G .
3. A single vertex in a graph G is a subgraph of G .
4. A single edge in G , together with its end vertices, is also a subgraph of G .

H is a subgraph of G .



Walk:

A walk in a graph is finite alternating sequence of vertices and edges beginning and ending with the vertices, such that edge is incident on the vertices preceding and following it.

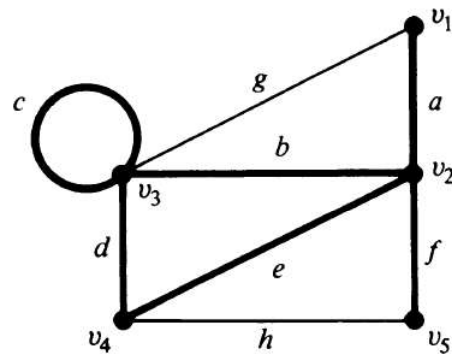
No edge appears (covered or traversed) more than once in a walk.

A vertex may appear more than once.

$v_1 a v_2 b v_3 c v_3 d v_4 e v_2 f v_5$ is a walk shown with heavy lines.

Vertices with which a walk begins and ends are called its *terminal vertices*.

It is possible for a walk to begin and end at the same vertex. Such a walk is called a *closed walk*. A walk that is not closed is called an *open walk*



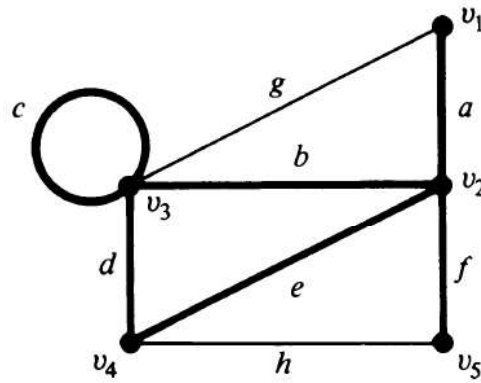
(a) An Open Walk

Path: An open walk in which no vertex appears more than once is called a *path*.
 In Fig. $v_1 a v_2 b v_3 d v_4$ is a path, whereas $v_1 a v_2 b v_3 c v_3 d v_4 e v_2 f v_5$ is not a path.

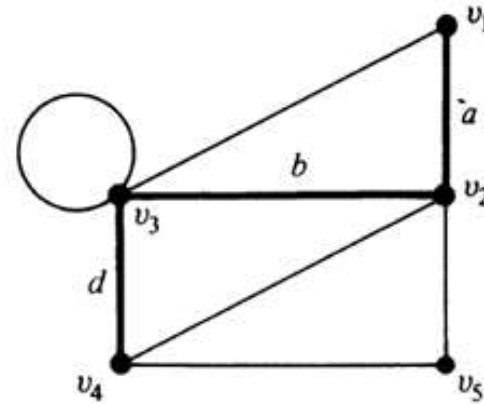
The number of edges in a path is called the *length of a path*.

An edge which is not a self-loop is a path of length one.

It should also be noted that a self-loop can be included in a walk but not in a path.



(a) An Open Walk



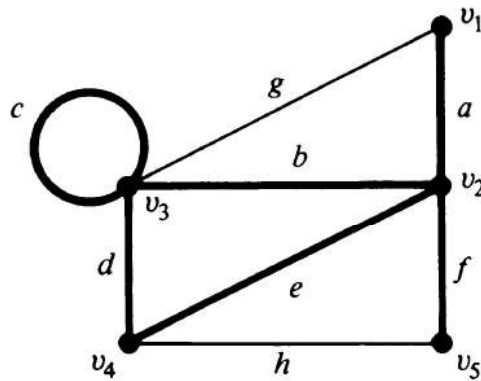
(b) A Path of Length Three

Circuit:

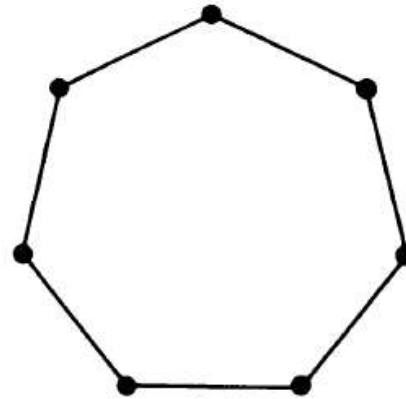
A closed walk in which no vertex (except the initial and the final vertex) appears more than once is called a *circuit*.

$v_2 b v_3 d v_4 e v_2$ is a circuit.

A circuit is also called a *cycle*, *elementary cycle*, *circular path*, and *polygon*.



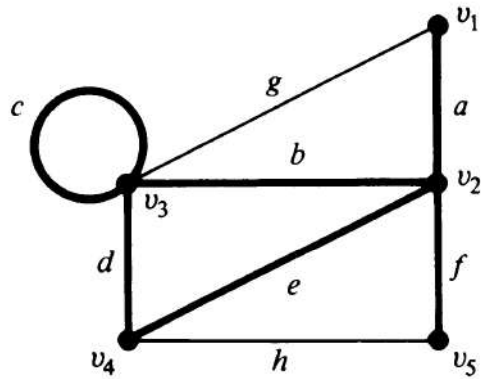
(a) An Open Walk



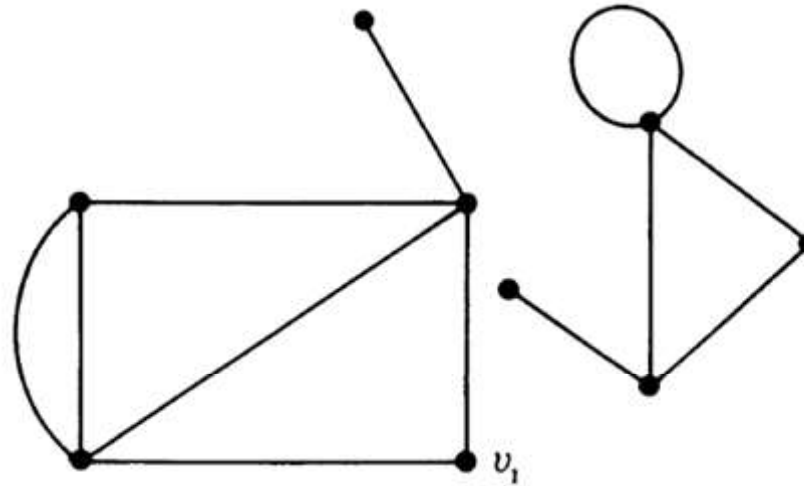
Three different circuits.

CONNECTED GRAPHS

A graph G is said to be *connected* if there is at least one path between every pair of vertices in G . Otherwise, G is *disconnected*.



connected graph



A disconnected graph with two components.

- A null graph of more than one vertex is disconnected.
- A disconnected graph consists of two or more connected graphs each of which is a connected subgraph is called a component.

Theorem: A graph G is disconnected if and only if its vertex set V can be partitioned into two nonempty subsets V_1 and V_2 such that there exist no edge in G whose one vertex is in subset V_1 and other is in V_2 .

Theorem: A simple graph (i.e., a graph without parallel edges or self-loops) with n vertices and k components can have at most $(n - k)(n - k + 1)/2$ edges.

Proof: Let the number of vertices in each of the k components of a graph G be $n_1, n_2, \dots, n_k$. Thus we have

$$n_1 + n_2 + \dots + n_k = n, \quad n_i \geq 1.$$

$$\sum_{i=1}^k n_i = n$$

$$\sum_{i=1}^k n_i - k = n - k$$

$$\sum_{i=1}^k (n_i - 1) = n - k$$

$$\left[\sum_{i=1}^k (n_i - 1)\right]^2 = n^2 + k^2 - 2nk$$

$$\sum_{i=1}^k (n_i - 1)^2 + 2 \sum_{i \neq j} (n_i - 1)(n_j - 1) = n^2 + k^2 - 2nk$$

$$\sum_{i=1}^k (n_i - 1)^2 \leq n^2 + k^2 - 2nk \quad \text{as} \quad \sum_{i \neq j} (n_i - 1)(n_j - 1) \geq 0 \text{ as each } n_i \geq 1.$$

$$(\sum_{i=1}^k n_i^2) - k + 2nk \leq n^2 + k^2 - 2nk$$

$$(\sum_{i=1}^k n_i^2) \leq n^2 - (2n - k)(k - 1) \quad \text{.....(1)}$$

We know maximum number of edges in i^{th} component is $\frac{n_i(n_i - 1)}{2}$

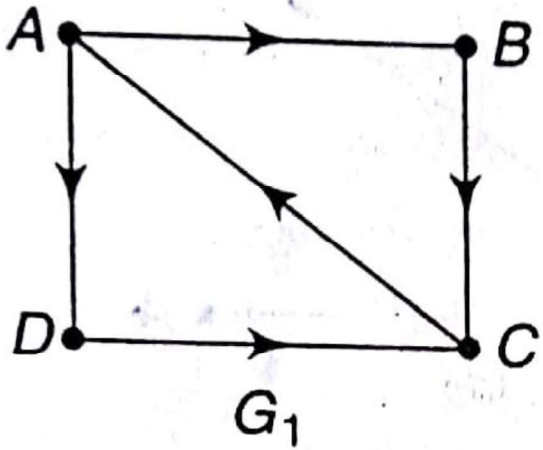
Maximum no. of edges in graph G are

$$\begin{aligned} \sum_{i=1}^k \frac{n_i(n_i - 1)}{2} &= \frac{1}{2} \sum_{i=1}^k n_i^2 - \frac{1}{2} \sum_{i=1}^k n_i \\ &\leq \frac{1}{2} (n^2 - (2n - k)(k - 1)) - \frac{n}{2} \\ &= \frac{1}{2} (n - k)(n - k + 1) \end{aligned}$$

Connectedness in directed graph:

A directed graph is said to be *strongly connected*, if there is a path from V_i to V_j and from V_j to V_i where V_i and V_j are any pair of vertices of the graph.

For a directed graph to be strongly connected, there must be a sequence of directed edges from any vertex in the graph to any other vertex.



G_1 is a strongly connected graph, as the possible pairs of vertices in G_1 are (A, B) , (A, C) , (A, D) , (B, C) , (B, D) and (C, D) and there is a path from the first vertex to the second and from the second vertex to the first in all the pairs.

DELETION OF VERTICES AND EDGES FROM THE GRAPH:

If v_i is a vertex in graph G , then $G - v_i$ denotes a subgraph of G obtained by deleting (i.e., removing) v_i from G .

Deletion of a vertex always implies the deletion of all edges incident on that vertex.

If e_i is an edge in G , then $G - e_i$ is a subgraph of G obtained by deleting e_i from G . Deletion of an edge does not imply deletion of its end vertices.

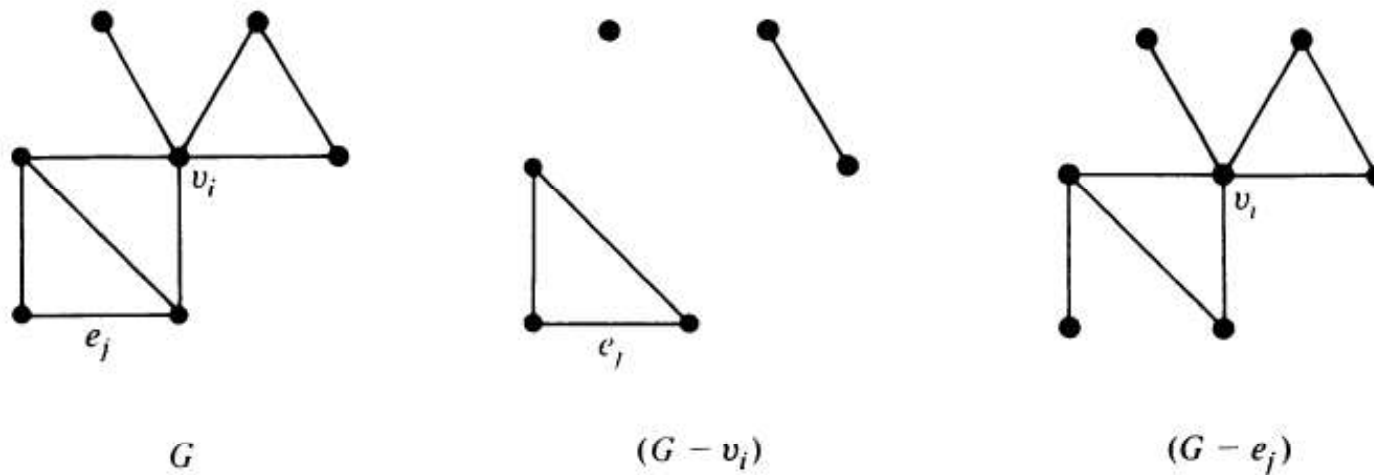


Fig. 2-15 Vertex deletion and edge deletion.

Cut Set:

In a connected graph G , a *cut-set* is a set of edges whose removal from G leaves G disconnected, provided removal of no proper subset of these edges disconnects G .

For instance, in Fig. 4-1 the set of edges $\{a, c, d, f\}$ is a cut-set.

There are many other cut-sets, such as $\{a, b, g\}$, $\{a, b, e, f\}$, and $\{d, h, f\}$. Edge $\{k\}$ alone is also a cut-set. The set of edges $\{a, c, h, d\}$, on the other hand, is *not* a cut-set, because one of its proper subsets, $\{a, c, h\}$, is a cut-set.

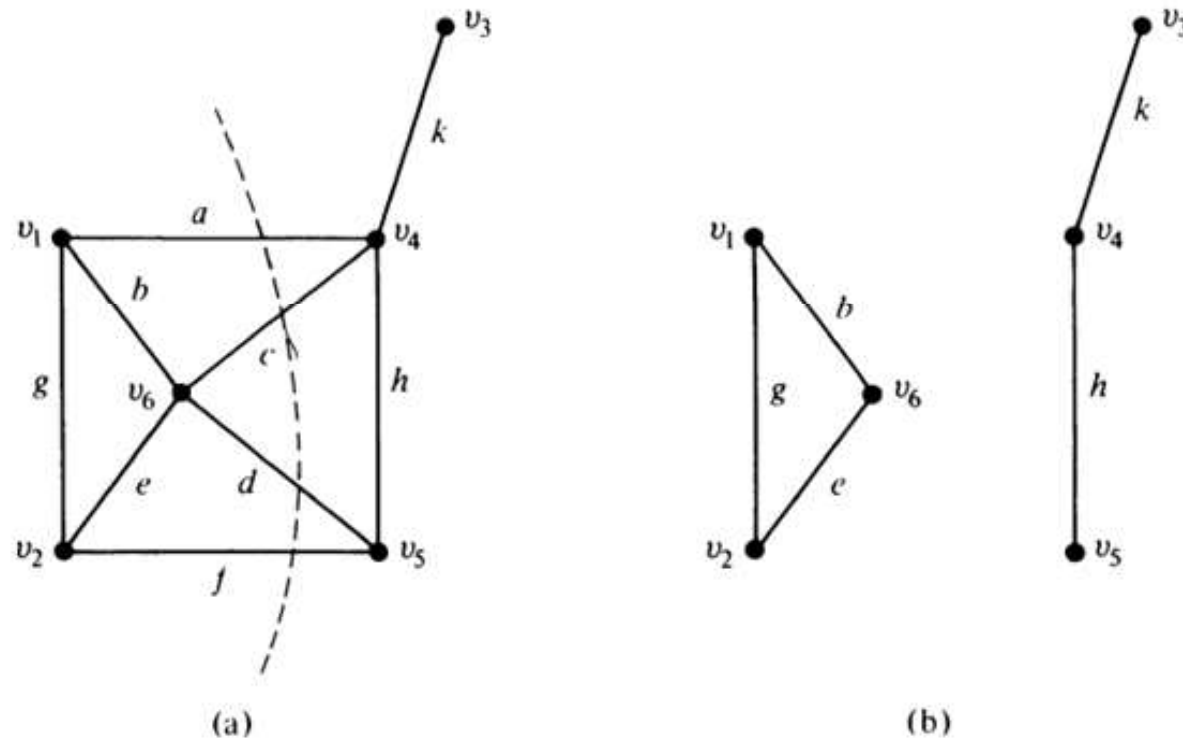


Fig. 4-1 Removal of a cut-set $\{a, c, d, f\}$ from a graph “cuts” it into two.

CONNECTIVITY

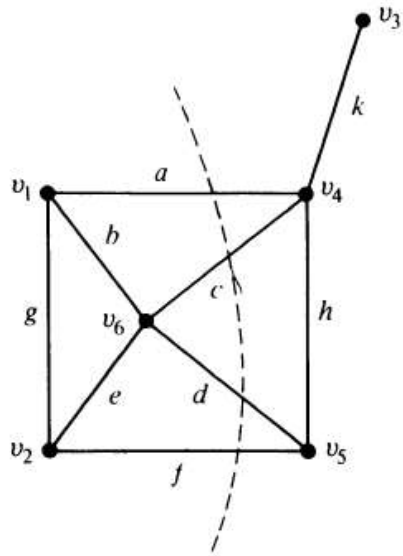
Edge Connectivity

Definition . Let G be a connected graph. The edge connectivity of G is the minimum number of lines (edges) whose removal results in a disconnected graph.

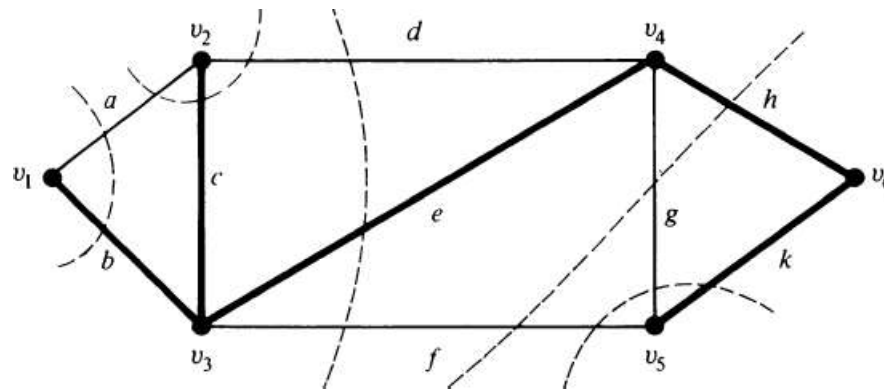
In other words

The number of edges in the smallest cut-set (i.e., cut-set with fewest number of edges) is defined as the *edge connectivity* of G .

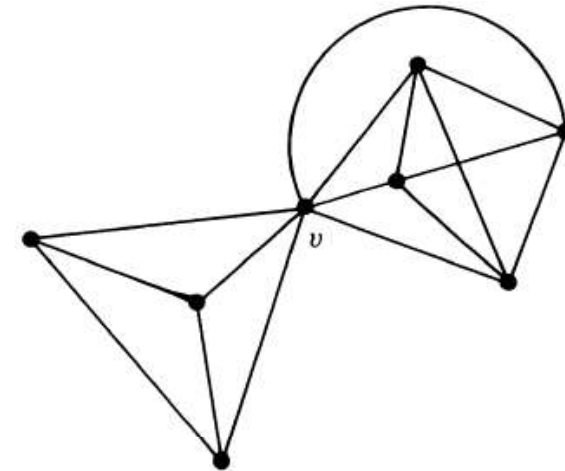
The edge connectivity of a connected graph G is denoted by $\lambda(G)$. If G is a disconnected graph, then $\lambda(G) = 0$.



Edge Connectivity - One



Edge Connectivity-Two



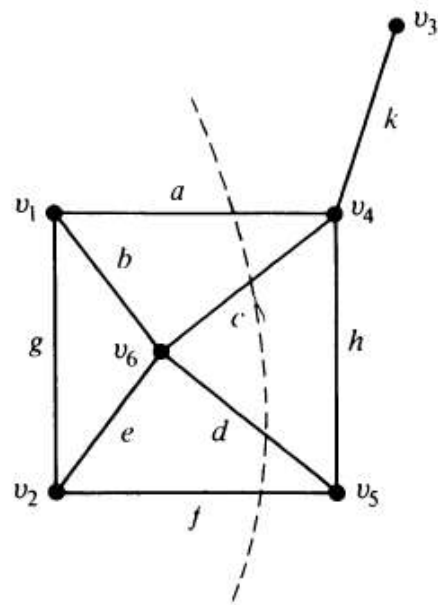
Edge Connectivity-Three

Vertex Connectivity

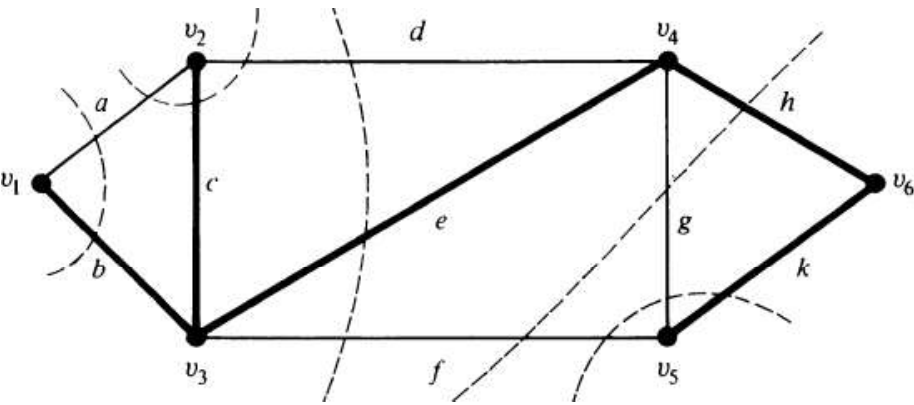
Definition Let G be a connected graph. The minimum number of vertices whose removal results in a disconnected or trivial graph is called the vertex connectivity of G .

The vertex connectivity of G is denoted by $k(G)$. If $k(G) = 1$, then G has a vertex v such that $G - v$ is not connected and the vertex v is called a cut vertex.

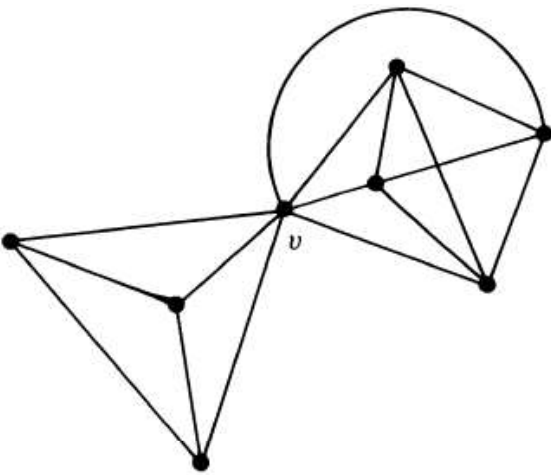
If $G = K_n$ the complete graph with n vertices then $k(G)$ is $n - 1$.



Vertex Connectivity - One



Vertex Connectivity - Two



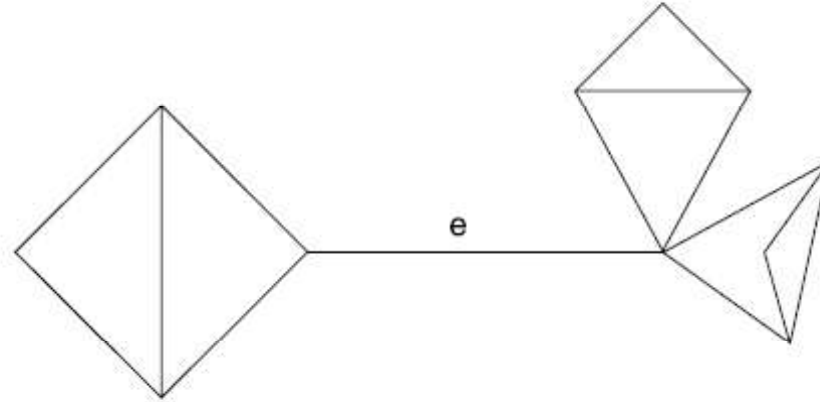
Vertex Connectivity - One

Theorem The vertex connectivity of a graph G is always less than or equal to the edge connectivity of G i.e., $k(G) \leq \lambda(G)$.

Theorem The edge connectivity of a connected graph G cannot exceed the minimum degree of G , i.e., $\lambda(G) \leq \delta(G)$.

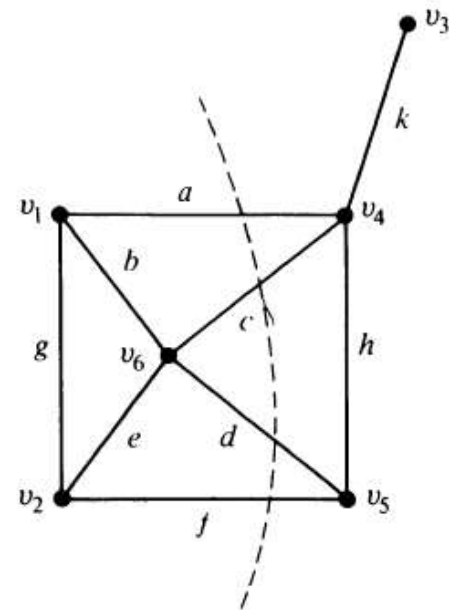
Theorem : For any graph G , $k(G) \leq \lambda(G) \leq \delta(G)$

Cut edge(Bridge): Let G be a connected graph. If e is an edge of G , such that $G-e$ is not connected, the edge e is called a cut edge (or bridge). In the graph the edge e is a cut edge.



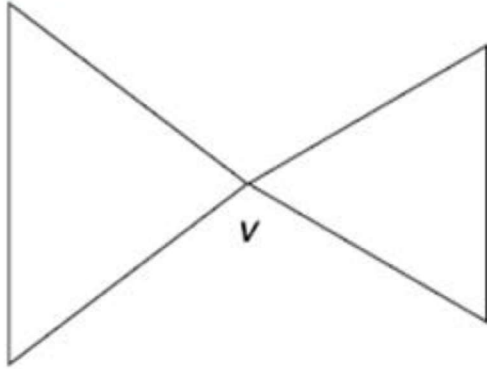
In other words if edge connectivity of a graph is one then that edge is called cut edge.

Ex: In the given graph edge connectivity is one and edge k is cut edge.



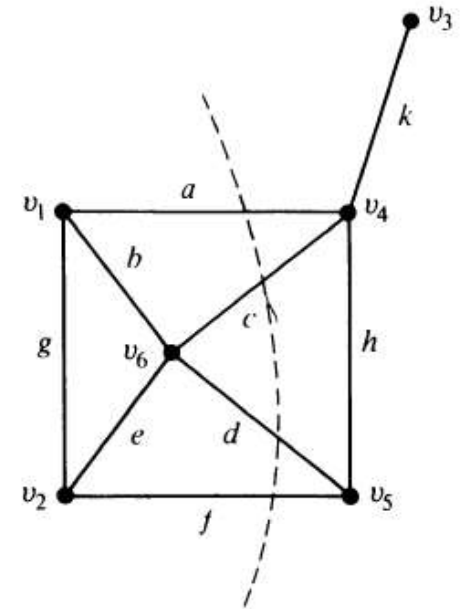
CUT VERTEX

Let G be a connected graph. If v is a vertex of G such that $G - v$ is not connected then, the vertex v is called a cut vertex.



If v is a cut vertex of G , then the removal of the vertex v increase the number of components in G . A cut vertex is also called a cut point.

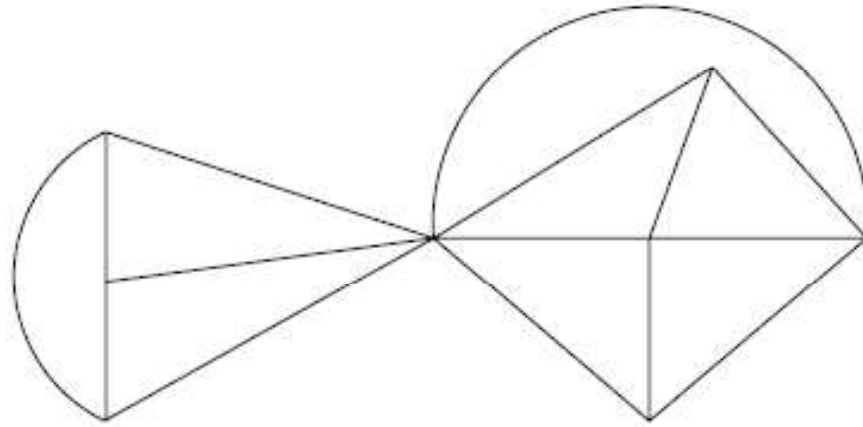
In other words if vertex connectivity is one then that vertex is called cut vertex.
Ex: In the given graph vertex connectivity is one and vertex v_4 is cut vertex.



Separable Graph:

A graph G is said to be separable if its vertex connectivity is one.

Example Find the edge connectivity and the vertex connectivity of the graph

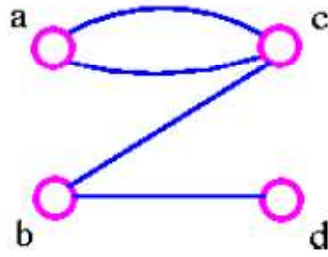


Solution: The minimum number of edges removal disconnects the graph is 3 and the minimum number of vertices required to disconnect the graph is 1.

$\therefore$ Edge connectivity $\lambda(G) = 3$

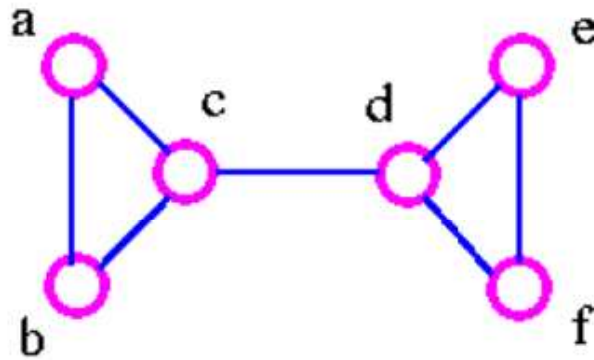
Vertex connectivity $k(G) = 1$.

Let us consider some examples:



The above graph can be split up into two components by removing one of the edges bc or bd. Therefore, edge bc or bd is a bridge.

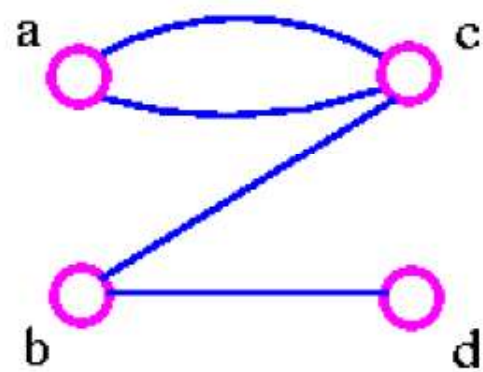
Thus the above graph has edge connectivity 1.



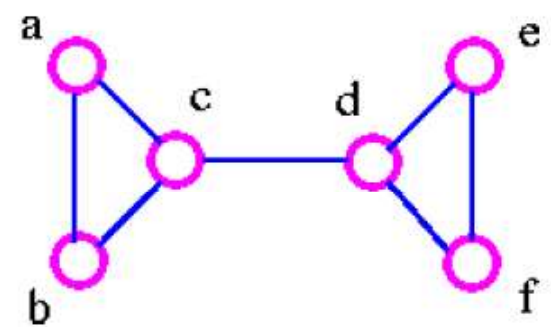
The above graph can be disconnected by removing a single edge, cd. Therefore, edge cd is a bridge.

Thus the above graph has edge connectivity 1.

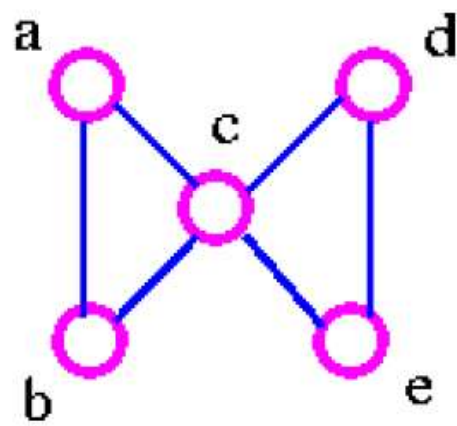
Let us consider some examples:



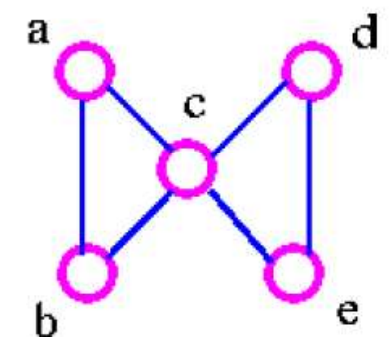
The above graph can be disconnected by removal of single vertex (either b or c). It has connectivity 1.



The above graph can be disconnected by removal of single vertex (either c or d). The vertex c or d is a cut-vertex. It has connectivity 1.



The above graph can be disconnected by removing just one vertex i.e., vertex c. The vertex c is the cut-vertex. It has connectivity 1.



The above graph cannot be disconnected by removing a single edge, but the removal of two edges (such as ac and bc) disconnects it. Thus the edge connectivity for the above graph is 2.

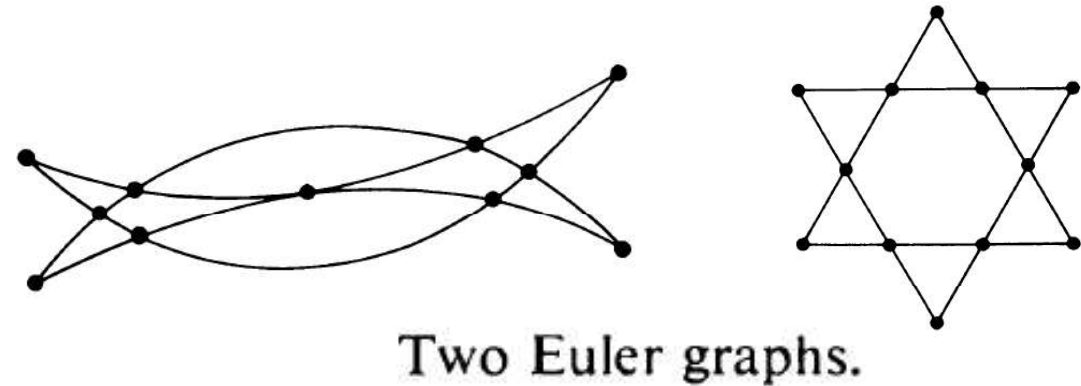
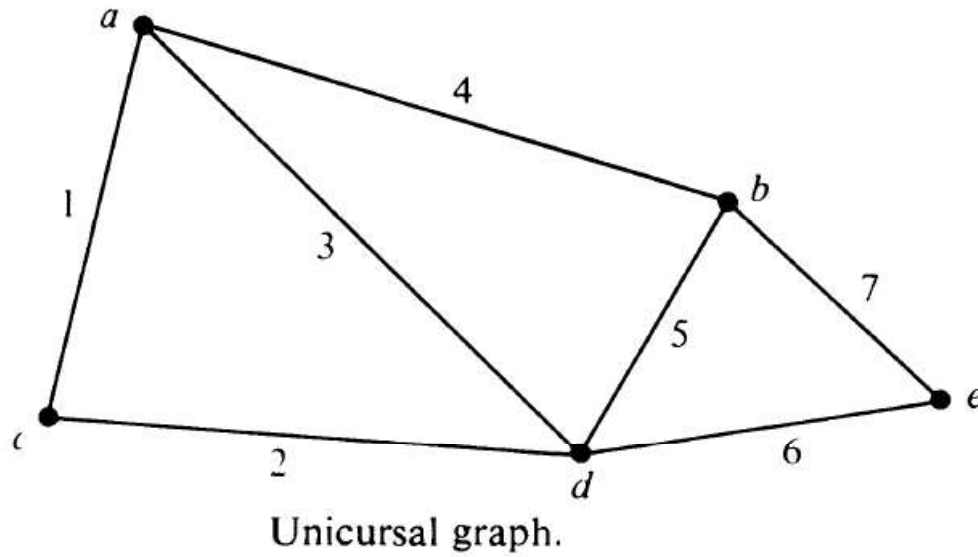
EULER GRAPHS

Eulerian line(or circuit): A closed walk of a graph G which includes every edge of the graph exactly once is called Eulerian line(or circuit) .

Eulerian Graph: A graph that contains Euler line is called Euler Graph.

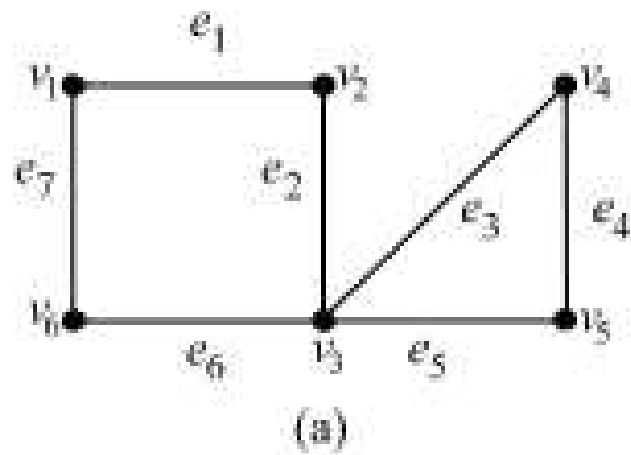
- Since the Euler circuit (or cycle) contains all the edges of the graph, therefore Euler graph is always connected.
- **Open Euler line**: An open walk that includes all edges of the graph G exactly once, is called open Euler line or unicursal line.

Unicursal graph: A graph that has unicursal line is called unicursal graph.

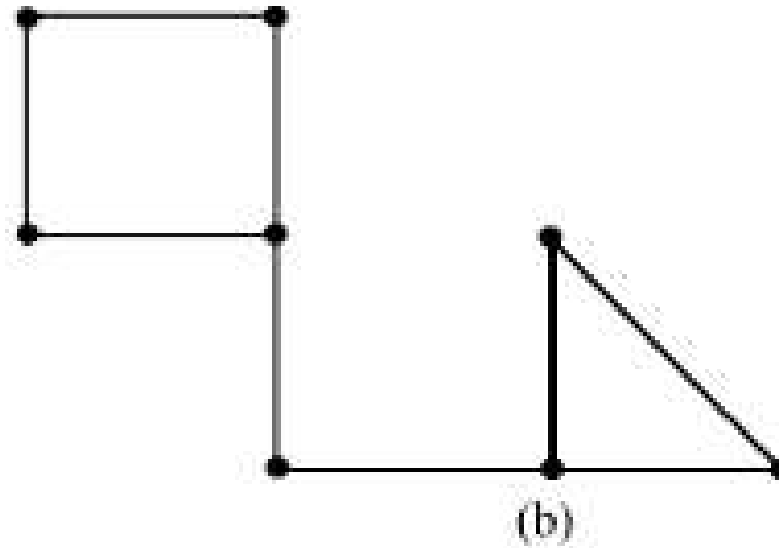


Two Euler graphs.

Example Consider the graph shown in Figure . Clearly, $v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_3 v_6 e_7 v_1$ in (a) is an Euler line, whereas the graph shown in (b) is non-Eulerian.



Eulerian Graph



Non-Eulerian Graph

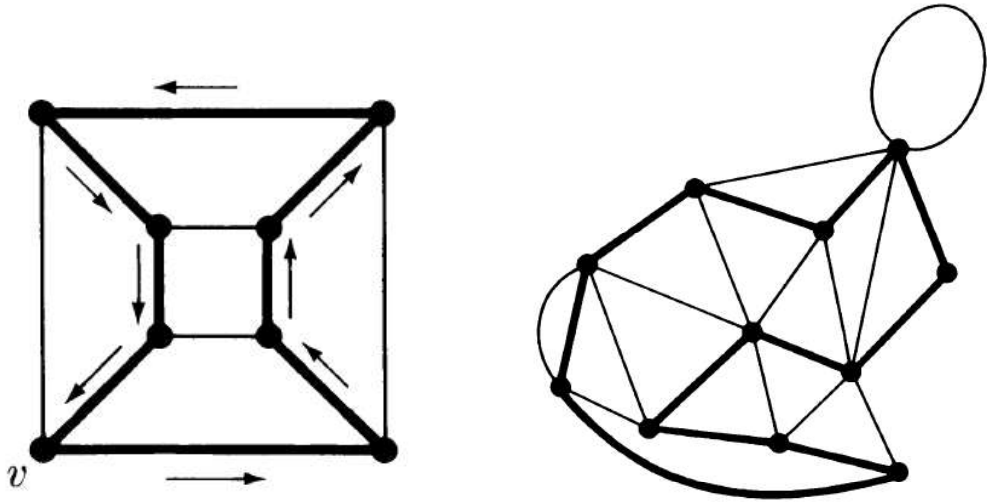
Theorem: A given connected graph G is an Euler graph if and only if all the vertices of G are of even degree.

HAMILTONIAN PATHS AND CIRCUITS

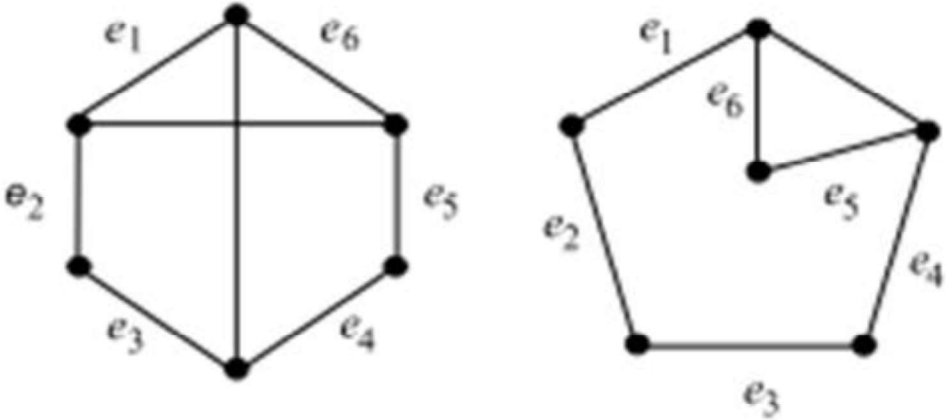
Hamiltonian Circuit:

A Hamiltonian circuit in connected graph G is defined as a closed walk that traverses every vertex of G exactly once, except the starting vertex at which walk terminates.

Hamiltonian circuit in a graph of n vertices consist of exactly n edges.



Hamiltonian circuits.



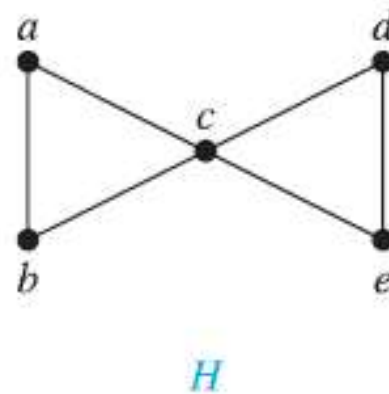
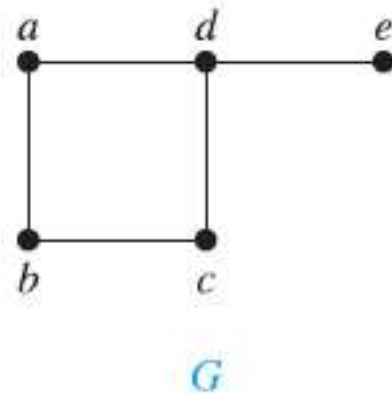
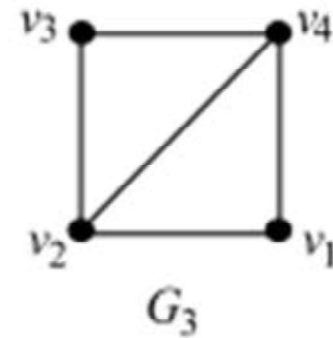
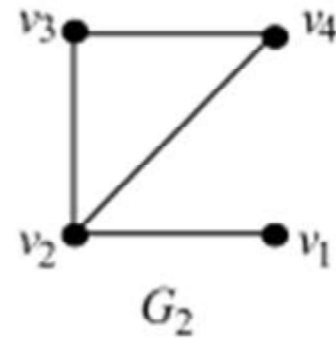
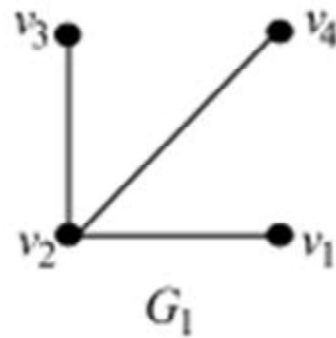
Hamiltonian Graphs

Hamiltonian Path: If we remove any one edge from a Hamiltonian circuit, we are left with a path. This path is called a *Hamiltonian path*. Clearly, a Hamiltonian path in a graph G traverses every vertex of G .

every graph that has a Hamiltonian circuit also has a Hamiltonian path. There are, however, many graphs with Hamiltonian paths that have no Hamiltonian circuits .

The length of a Hamiltonian path (if it exists) in a connected graph of n vertices is $n - 1$.

For example, in Figure, G_1 has no Hamiltonian path, and so no Hamiltonian cycle;
 G_2 has the Hamiltonian path $v_1v_2v_3v_4$, but has no Hamiltonian cycle,
 while G_3 has the Hamiltonian cycle $v_1v_2v_3v_4v_1$.



Two Graphs That Do Not Have a Hamilton Circuit.

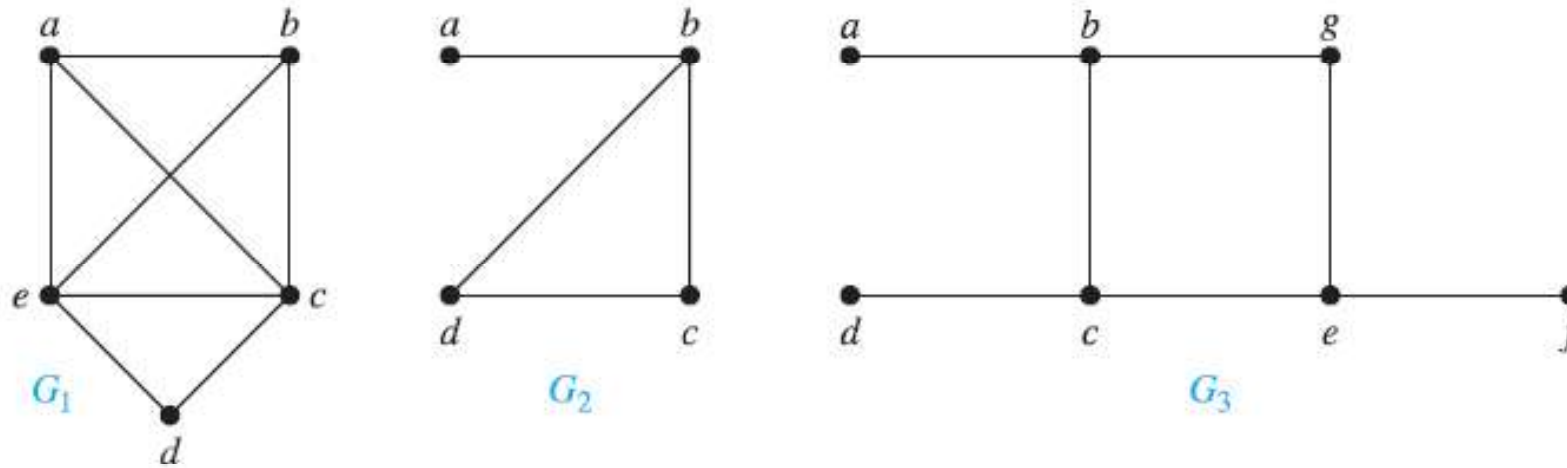
Few Properties of Hamiltonian Graphs:

- Path obtain by deleting any edge from a Hamiltonian circuit is called Hamiltonian path.
- Hamiltonian circuit contains a Hamiltonian path, but a graph containing a Hamiltonian path may not have Hamiltonian circuit.
- A complete graph K_n will always have a Hamiltonian circuit when, $n \geq 3$.
- A given graph may contain more than one Hamiltonian circuit.

EXAMPLE

Which of the simple graphs in Figure have a Hamilton circuit or, if not, a Hamilton path?

Solution: G_1 has a Hamilton circuit: a, b, c, d, e, a . There is no Hamilton circuit in G_2 (this can be seen by noting that any circuit containing every vertex must contain the edge $\{a, b\}$ twice), but G_2 does have a Hamilton path, namely, a, b, c, d . G_3 has neither a Hamilton circuit nor a Hamilton path, because any path containing all vertices must contain one of the edges $\{a, b\}$, $\{e, f\}$, and $\{c, d\}$ more than once. ▶



Three Simple Graphs.